The Paleomagnetism of the Shawa Ijolite

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Abstract. Oriented cores have been drilled at seven sites in ijolite exposures of the Shawa carbonatite complex, Southern Rhodesia. The ijolite has been dated at 209 ± 16 m.y. from Rb → Sr decay. The directions of natural remanent magnetization are not consistent between sites. The ijolite proved difficult to treat by partial demagnetization in alternating fields, as instability of the direction of remanent magnetization tended to develop with demagnetizing fields only slightly larger than those required to remove isothermal magnetizations. Nevertheless, consistent directions of remanent magnetization were secured from five of the seven sites, after alternating-field treatment. On the assumption of a geocentric dipole field, the mean south magnetic pole position is 64.2°S, 85.6°E, with radius of 95 per cent confidence 14.1°. This is in excellent agreement with other Mesozoic pole positions from Africa.

The Shawa carbonatite complex. This small ring complex has been intruded into the basement granite of eastern Mashonaland, Southern Rhodesia, at latitude 19°13′S, longitude 31° 42′E. The related complex at Dorowa lies 18 km to the north-northeast of Shawa. The geology of these intrusions has been described by *Johnson* [1961], and a simplified version of Johnson's map of Shawa is given (Figure 1). The intrusion is about 5½ km in diameter and is essentially a plug of almost completely serpentinized dunite partly enclosed by arcuate bodies of ijolite. Around these rocks lies a zone of fenite, grading outward into the granitic country rocks. A central ring-dike of carbonatite is intruded into the serpentinite. The carbonatite and fenite rings form concentric ranges of hills. Within the carbonatite ring, and between it and the fenite ring, the country is relatively flat over the serpentinite.

Shawa is of paleomagnetic interest because biotite from coarse ijolite of the western ijolite limb has been dated by $Rb^{87} \rightarrow Sr^{87}$ decay [*Nicolaysen et al.*, 1962]. The age found was 209 ± 16 m.y. using decay constant 1.386×10^{-11} yr^{-1} for Rb^{87}. Nicolaysen and his collaborators pointed out that the standard error of 16 m.y. gives 99 per cent confidence limits of ±40 m.y. and that the Shawa intrusion might therefore be post-Stormberg in age, i.e. not older than lower Jurassic. This was argued to account for the supposed Stormberg age of dolerites in the area, which Johnson had shown to be older than the Shawa intrusion. As these dolerites are now known, from a combination of paleomagnetic and $K^{40} \rightarrow Ar^{40}$ dating studies, to be not less than 1600 m.y. old [*McElhinny and Opdyke,* 1964], there is no reason why the measured values of 209 ± 16 m.y. for the age of the Shawa ijolite should not be accepted.

Of the rock types available in the Shawa complex, only ijolite offers much prospect of successful paleomagnetic study. The essential minerals are nepheline and pyroxene, with biotite, iron ore, and apatite as common accessories. The texture of the rock is extremely variable; it ranges from fine-grained ijolite in the chilled zone near the inner contact with the serpentinite to very coarse-grained pegmatite ijolite.

Sampling and measurement. Oriented cores were drilled at seven sites in the ijolite during September 1963. The sites are shown in Figure 1. There are no developed streams in the ijolite, and outcrops under active erosion could not be found. For this reason, long oriented cores (2 to 3 m) were taken at five sites (numbered 2 to 6, Figure 1) by means of the sampling drill described by *Gough and Opdyke* [1963]. At sites 1 and 7 five oriented cores 20 to 50 cm long were secured by means of a Graham-Keiller type drill [*Graham and Keiller,* 1960]. All the cores were oriented relative to the

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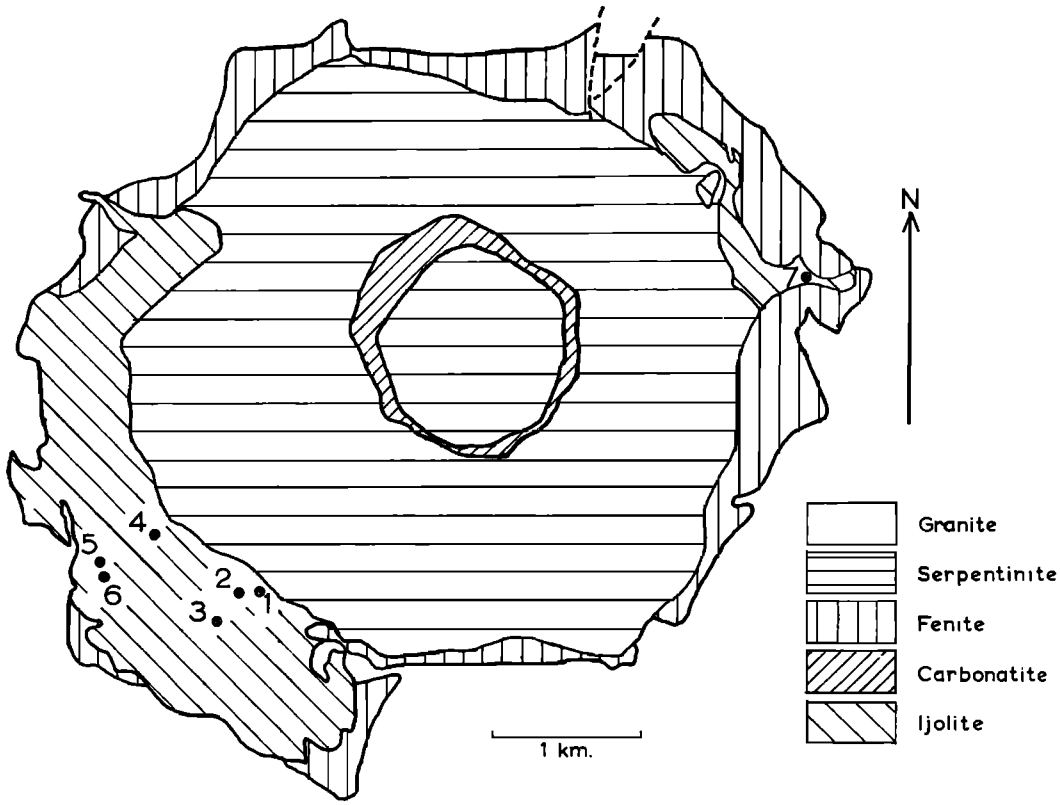


Fig. 1. The Shawa carbonatite complex and the sampled sites (after R. L. Johnson).

sun's azimuth by a technique described by *Gough and Opdyke* [1963]. The remanent magnetizations of specimens cut from these cores were measured with a spinner magnetometer [*Gough*, 1964], and stable directions of remanent magnetization were isolated by partial demagnetization of the specimens in alternating fields by means of apparatus described by *McElhinny and Gough* [1963].

The texture of the rock varied widely between the sites and also, in some cases, over distances of a few centimeters along a core. Site 1 was in the chilled zone near the serpentinite contact, and the rock is of fine-to-medium grain. At site 7 the ijolite, which in the northeast intrusion is richer in biotite than in the southwest limb, is of medium grain size. At the other sites the grain size varied irregularly from medium to very coarse. No relation was observed between grain size and any relevant magnetic property of the rock.

Paleomagnetic results. At no site did the direction of the natural remanent magnetization (NRM) remain unchanged after alternating-field treatment. The directions of NRM were scattered at sites 1 and 7, sampled with the small drill, and became fairly well grouped after treatment. At sites sampled with the large drill the specimens were cut from a single core which was 2 to 3 m long. The directions of NRM of such specimens were not widely scattered; in some instances they were closely grouped. Upon treatment in alternating fields of maximum amplitude 100 to 300 gauss, the directions of magnetization moved to give a new group in a quite different direction. This type of behavior is illustrated in Figure 2, which shows directions of NRM and of remanent magnetization after treatment at 150 gauss, for specimens from the long core drilled at site 2. In this case NRM directions were so closely grouped that the Fisher precision parameter k [*Fisher*, 1953] was smaller, though not significantly so, after demagnetization than before

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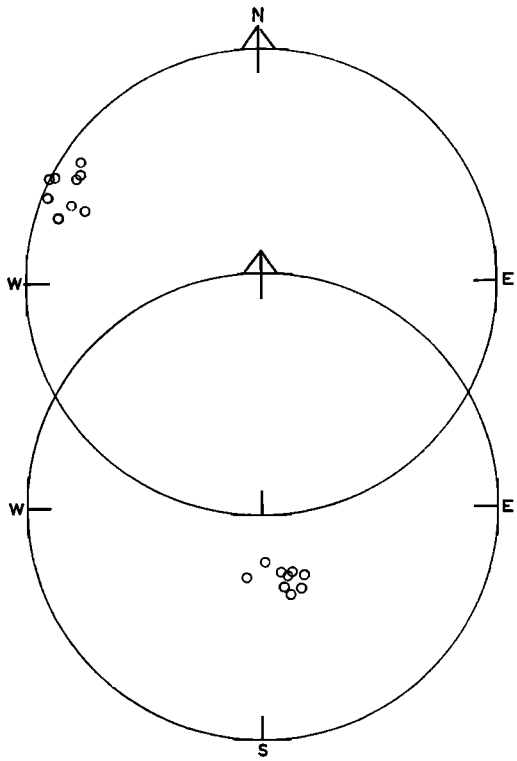


Fig. 2. Directions of NRM (upper figure) and remanent magnetization after treatment in 150-gauss alternating field (lower figure) at site 2. South-seeking directions down; stereographic projection.

(Table 1). The mean direction of magnetization for this core moved through 109° during the partial demagnetization. Similar displacement of the direction of magnetization of a whole long core has been observed in the Lupata volcanics [*Gough and Opdyke*, 1963] and in the Marangudzi ring complex [*Gough et al.*, 1964]. It is probable that it occurs when a large lightning current flows at a distance such that its magnetic field varies little in direction along a core 2 or 3 m long. In such cases the NRM directions, although well grouped at each site, disagree widely between sites. This can be seen in the NRM parameters for the Shawa long-core sites 2, 3, and 4 (Table 1). After partial demagnetization in alternating fields, the mean directions given by these sites agreed, subject to the usual dispersion caused by secular variation.

Many igneous rocks, upon partial demagnetization in progressively higher alternating magnetic fields, first show changes in the direction of the remanent magnetization and then show no further changes of direction after the field passes a certain value. In such cases the direction of a stable magnetization of high coercivity is readily isolated from components, probably isothermal in origin, of lower coercivity. If the directions given by specimens from a site then agree, and if, further, the site mean directions agree, there is a strong presumption that the direction of the original thermoremanent magnetization has been isolated. Examples of this 'ideal' behavior are given by *As and Zijderveld* [1958], *Irving et al.* [1961], *McElhinny and Gough* [1963], and *Gough and Opdyke* [1963]. Other igneous rocks show progressive changes of magnetization as the alternating field rises to a certain value, and

TABLE 1. Mean Directions of Magnetization and Site Pole Positions from Five Sites

				M_N					M_{ac}					Site South Pole	
Site	N_c	N	E	D, deg	I, deg	R	k	B_{ac}, gauss	D, deg	I, deg	R	k	α_{95}, deg	Lat. S, deg	Long. E, deg
1	5	10	0	(24.6)	(−10.0)	2 903	(1 9)	200–400	349.6	−48 9	9.310	13 0	13.9	75.8	71.2
2	1	9	0	116.3	−5 0	8.944	142	150	344.0	−54.5	8.896	76.7	5.9	68.8	70.3
3	1	10	0	16.4	+3.9	9.742	34.8	300	341.7	−55.5	9.759	37.3	8.0	66.7	71 8
4	1	13	0	67 3	−27.6	11.637	8.80	150	332.2	−39.9	12.406	20.2	9.5	63.9	109.1
7	5	10	4	(352.0)	(−21.6)	2.903	(1.9)	150	308.2	−54.8	5.664	14.9	18.0	41.8	91.0

N_c = number of cores drilled.
N = number of specimens measured.
E = number of specimens excluded in calculating results.
M_N denotes natural remanent magnetization.
M_{ac} denotes magnetization after treatment in alternating field of maximum amplitude B_{ac}.
D = declination of remanent magnetization.
I = inclination of remanent magnetization.
R = resultant of N unit vectors except for M_N at sites 1 and 7, where R = resultant of N_c unit vectors.
k = $(N - 1)/(N - R)$ = estimate of Fisher's κ parameter.
α_{95} = semiangle of cone of 95 per cent confidence for mean direction.

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then exhibit instability of the direction of remanent magnetization [*Irving et al.*, 1961; *Gough et al.*, 1964]. If instability occurs before the directions within the site converge to form a group, nothing can be done with the specimens from that site for paleomagnetic purposes. Sometimes, however, the directions for specimens within a site begin to form a group before instability occurs, and it becomes possible to find the direction of a magnetization of higher coercivity by demagnetizing all the specimens by means of fields just too small to cause instability. Often several successive values of demagnetizing field must be applied to each specimen, since the onset of instability varies between specimens.

Most of the ijolite specimens from Shawa showed one of the two latter types of behavior. Specimens from two long cores (sites 5 and 6) of coarse-grained ijolite gave no result because their directions of magnetization became unstable before a group could be formed. Most specimens from the other five sites had magnetizations whose directions formed groups at certain demagnetizing field values but became unstable at higher fields.

Numerous specimens were treated in progressively higher alternating fields in order to establish as closely as possible the direction of the more stable component. It was observed that in nearly all cases the onset of instability was accompanied by a characteristic sequence of events. Reference should be made to the description of the spinner magnetometer and its operation by *Gough* [1964].

1. The corrections to pairs of angles to make their sums 360° became variable. If the nulling fields were switched off, large second-harmonic components appeared on the oscilloscope.

2. Second-harmonic components became noticeable even with the geomagnetic field approximately null.

3. Corrections to pairs of angles became extremely variable, and intersections of the radius and two arcs of the stereographic net became increasingly poor.

4. Instability of the direction of remanent magnetization occurred simultaneously with events (2) and (3). An extreme example of instability is discussed by *Gough et al.* [1964, Figure 4].

Gough [1964] has pointed out that events (1), (2), and (3) are to be expected when the specimen has low remanent magnetization J but high susceptibility K, and consequently low Koenigsberger ratio $Q = J/KT$ (T = geomagnetic field intensity). It is possible that instability of the direction of remanent magnetization is associated with a minimum possible value of Q for the rock. This possibility will be investigated. Such a possibility would lend support to the hypothesis of *Irving et al.* [1961] that instability occurs when directions of saturation magnetization of N domains have been randomized so that the resultant remanent intensity J is the minimum possible sum of N randomly oriented vectors.

The mean directions of natural remanent magnetization at five sites are given in Table 1, and also the directions of magnetization after alternating-field treatment. For a minority of specimens it was possible to show that little further change of direction resulted on demagnetization in increasing fields and that a stable component had therefore been isolated. For the majority of specimens this was not possible because of the onset of instability. However, alternating-field treatment gave rise to a group of directions of magnetization for specimens from each site. Moreover, the site mean directions were found to agree. It is therefore believed that the data represent reasonably well the direction of the original thermoremanent magnetization. The results are not as accurate as those from basic igneous rocks like gabbros or dolerites, because of the magnetic properties of the ijolite.

Pole position. As shown in Table 1, all sites are normally magnetized. The cones of 95 per cent confidence for the site mean directions are shown in Figure 3. On the assumption of a geocentric dipole field, the south paleomagnetic pole position is latitude 64.2°S, longitude 85.6°E, $K = 30$, $A_{95} = 14.1°$. The relatively large value of A_{95} reflects the lower precision of these results from ijolite as compared with results from more normal basic igneous rocks. The pole position is 3.3° from the north pole position given by the slightly younger ring complexes at Marangudzi and the Mateke Hills [*Gough et al.*, 1964].

It would be unwise, on results from such a small intrusion, to attempt to answer the ques-

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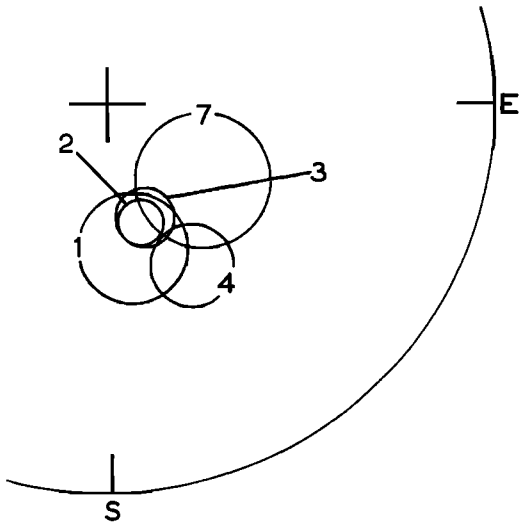


Fig. 3. Cones of 95 per cent confidence for mean directions of magnetization at five sites. South-seeking directions down; stereographic projection.

tion whether the times of magnetization at the sites cover a sufficient time interval to yield a mean pole position free from displacement by secular variation. Sites were distributed across the width of the main ijolite intrusion in an attempt to achieve this objective.

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